

A Sublinear Two-Sided Near-Additivity Counterexample

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Status. Candidate full counterexample, likely valid, for human review.

1 Source question

Briceño and Iommi prove that if a sequence $(f_n)_n \subseteq \mathcal{C}_b(X)$ satisfies

$$\|f_{n+m} - f_n - f_m \circ T^n\|_\infty \leq C_n + C_m \quad (m, n \in \mathbb{N}) \quad (1)$$

for a nondecreasing nonnegative sequence such that $\sum_{n \geq 1} C_n/n^2 < \infty$, then $(f_n)_n$ is asymptotically additive [1, Proposition 5.1]. Their Question 5.2 asks whether the summability assumption can be replaced by only

$$0 \leq C_n \leq C_{n+1}, \quad \frac{C_n}{n} \longrightarrow 0.$$

The question occurs on PDF page 27.

Remark 9. Proposition 5.1 combined with Proposition 3.7 and Proposition 3.11 guarantee that every sequence $(f_n)_n$ that satisfies Equation (7) is Riesz-almost additive (of constant type), and thus almost additive. Notice also that if $(C_n)_n$ is constant in Proposition 5.1, the same conclusion holds, generalizing [38, Proposition 2.1].

Question 5.2. *If $(C_n)_n$ is a non-decreasing sequence of positive real numbers such that $\sum_{n \geq 1} \frac{C_n}{n^2} < \infty$, then $\lim_{n \rightarrow \infty} \frac{C_n}{n} = 0$. Is it possible to weaken the assumptions on $(C_n)_n$ in Proposition 5.1 to just $0 \leq C_n \leq C_{n+1}$ and $\lim_{n \rightarrow \infty} \frac{C_n}{n} = 0$? An analogue question in the setting of nearly sub-additive sequences of real numbers is answered negatively in [17].*

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2 Result

Theorem 1. *The proposed weakening in Question 5.2 is false. More precisely, there are a compact dynamical system (X, T) , a sequence $(f_n)_n \subseteq \mathcal{C}_b(X)$, and a sequence $(C_n)_{n \geq 0}$ such that:*

1. $C_0 = 0$, and $(C_n)_{n \geq 1}$ is positive, nondecreasing, and even subadditive;
2. $C_n/n \rightarrow 0$;

3. (1) holds for all $m, n \geq 0$;
4. $(f_n)_n$ is not asymptotically additive.

In fact, the failure in the last item has normalized distance exactly one from the additive sequences.

3 Proof intuition

On a one-point system, asymptotic additivity is simply convergence of f_n/n . We therefore want a bounded, nonconvergent slope $h(n) = f_n/n$ which changes so slowly that $f_n = nh(n)$ is additive up to a sublinear error. The choice

$$h(x) = \sin(\log \log(x + e^4))$$

does exactly this: its phase tends to infinity, so it keeps oscillating, while its derivative is bounded by $((x + e^4) \log(x + e^4))^{-1}$. For two comparable summands this derivative directly gives an $O(n/\log n)$ defect. If one summand is much smaller, its coefficient absorbs the larger change in phase. Splitting at $m = \sqrt{n}$ makes this uniform and yields the explicit error $C_n = 15n/\log(n + e^4)$.

4 Proof

Put $B = e^4$ and define, for $x \geq 0$,

$$q(x) = \log \log(x + B), \quad h(x) = \sin q(x).$$

Then

$$|h'(x)| \leq q'(x) = \frac{1}{(x + B) \log(x + B)}. \quad (2)$$

Take $X = \{*\}$ with $T(*) = *$. Set

$$f_0(*) = 0, \quad f_n(*) = nh(n) \quad (n \geq 1),$$

and

$$C_0 = 0, \quad C_n = \frac{15n}{\log(n + B)} \quad (n \geq 1). \quad (3)$$

Every f_n is of course bounded and continuous.

We first verify the error estimate. By symmetry it is enough to consider $1 \leq m \leq n$. Algebra and (2) give

$$\begin{aligned} |f_{n+m} - f_n - f_m| &\leq n|h(n+m) - h(n)| + m|h(n+m) - h(m)| \\ &\leq n \int_n^{n+m} \frac{dx}{(x + B) \log(x + B)} + m \int_m^{n+m} \frac{dx}{(x + B) \log(x + B)}. \end{aligned} \quad (4)$$

The first integral term is at most

$$\frac{nm}{(n + B) \log(n + B)} \leq \frac{n}{\log(n + B)}. \quad (5)$$

Denote the second term by $E_{m,n}$. We claim that

$$E_{m,n} \leq \frac{14n}{\log(n+B)}. \quad (6)$$

Suppose first that $m \leq \sqrt{n}$. Since $m \leq n$ and $\log B = 4$,

$$\begin{aligned} E_{m,n} &= m \log \left(\frac{\log(n+m+B)}{\log(m+B)} \right) \\ &\leq \frac{m}{4} \log(2(n+B)). \end{aligned}$$

Here we used $\log y \leq y$ for $y > 0$. Since $n+B \leq (B+1)n < e^5 n$ and $2(n+B) < e^6 n$, while $\log n \leq 2n^{1/4}$ for $n \geq 1$, it follows that

$$\begin{aligned} \frac{E_{m,n} \log(n+B)}{n} &\leq \frac{\log(n+B) \log(2(n+B))}{4\sqrt{n}} \\ &\leq \frac{(5+2n^{1/4})(6+2n^{1/4})}{4\sqrt{n}} \\ &= \frac{30}{4\sqrt{n}} + \frac{22}{4n^{1/4}} + 1 \leq 14. \end{aligned}$$

Now suppose that $m > \sqrt{n}$. Using $\log(1+u) \leq u$ twice,

$$\begin{aligned} E_{m,n} &= m \log \left(1 + \frac{\log(1+n/(m+B))}{\log(m+B)} \right) \\ &\leq \frac{m}{\log(m+B)} \log \left(1 + \frac{n}{m+B} \right) \leq \frac{n}{\log(m+B)}. \end{aligned}$$

Moreover $(m+B)^2 > n+B$, so $\log(n+B) < 2\log(m+B)$. Consequently

$$E_{m,n} \leq \frac{2n}{\log(n+B)},$$

which is stronger than (6). Combining (4)–(6) yields

$$|f_{n+m} - f_n - f_m| \leq \frac{15n}{\log(n+B)} = C_n \leq C_n + C_m.$$

The cases $m = 0$ or $n = 0$ are identities, so (1) holds for all nonnegative m, n .

Next consider the hypotheses on the errors. The function $x \mapsto x/\log(x+B)$ is strictly increasing on $[0, \infty)$, because its derivative has the sign of

$$\log(x+B) - \frac{x}{x+B} > 4 - 1 > 0.$$

Thus (C_n) is nondecreasing. Also

$$\frac{C_n}{n} = \frac{15}{\log(n+B)} \longrightarrow 0.$$

It is even subadditive: since $x \mapsto 1/\log(x + B)$ decreases,

$$\frac{n + m}{\log(n + m + B)} \leq \frac{n}{\log(n + B)} + \frac{m}{\log(m + B)}.$$

For comparison with Proposition 5.1, the excluded summability condition does indeed fail, since

$$\sum_{n \geq 1} \frac{C_n}{n^2} = 15 \sum_{n \geq 1} \frac{1}{n \log(n + B)} = \infty.$$

It remains to show failure of the conclusion. Choose integers n_k^+ and n_k^- nearest, respectively, to

$$\exp(\exp(\frac{\pi}{2} + 2\pi k)) - B, \quad \exp(\exp(\frac{3\pi}{2} + 2\pi k)) - B.$$

Since $q'(x) \rightarrow 0$, rounding changes q by $o(1)$. Hence

$$h(n_k^+) \rightarrow 1, \quad h(n_k^-) \rightarrow -1.$$

Thus $f_n/n = h(n)$ does not converge.

More exactly, every $g \in \mathcal{C}_b(X)$ is a scalar λ , and $S_n g = n\lambda$. Therefore

$$\limsup_{n \rightarrow \infty} \frac{\|f_n - S_n g\|_\infty}{n} = \limsup_{n \rightarrow \infty} |h(n) - \lambda| \geq \max\{|1 - \lambda|, |-1 - \lambda|\} \geq 1.$$

Equality is attained at $\lambda = 0$. Hence the infimum in the definition of asymptotic additivity is exactly one, rather than zero. This proves Theorem 1. $\square$

5 Verification, scope, and novelty

The proof uses no external theorem beyond the source definitions. The included finite-range verifier exhaustively checks the defect for all $1 \leq m \leq n \leq 800$, checks selected logarithmic-scale pairs up to $n = 10^{12}$, and checks finite-range monotonicity and subadditivity of (C_n) . These computations are not used in the proof.

The bounded novelty search on 22 July 2026 covered the run's four lightweight indexes, the exact arXiv id and question wording, and web/arXiv searches for the phrases **C_n + C_m, asymptotically additive**, and **sublinear nearly additive**, with close variants. No separate paper explicitly resolving Question 5.2 was found. The source notes only that an analogous one-sided nearly-subadditive scalar question was answered negatively by Füredi. This packet does not rely on that construction. Novelty is not certified beyond the bounded search.

The counterexample settles the yes/no weakening in Question 5.2. It does not seek the sharp additional regularity on (C_n) that would restore the conclusion. Human review should focus on the two cases in (6); the one-point reduction and the final oscillation are otherwise direct.

References

- [1] R. Briceño and G. Iommi, *Ergodic theorems for set maps under weak forms of additivity*, arXiv:2501.14581v1 (2025).